Lecture 1: Fault Tolerant System Design

Seyedhamidreza Mousavi

Outlines

- Why fault tolerant system design?
- Course Focus
  - Objectives
  - Overview
- Other Course Information

Why FT system design

- Electronic systems are an indispensable part of all our lives.

Why FT system design (Cont'd)

- Electronic systems are an indispensable part of all our lives.
- Malfunctions in these systems have consequences ranging from annoying computer crashes, loss of data and services, to *financial* and *productivity* losses, or even loss of *human life*.
  - For example, in 2009, a glitch in a single circuit board in the air-traffic control system resulted in hundreds of flights being canceled or delayed.

Why FT system design (Cont'd)

- Malfunctions may be caused by hardware or software failures, design errors, malicious attacks, or incorrect human interactions.
  - Such impacts continue to increase as systems become more complex, interconnected, and pervasive.

- *Fault Tolerant System Design* is required to ensure that future systems *perform correctly* despite rising levels of complexity and increasing disturbances.

Focus-Objectives

- understanding fault tolerance
  - faults and their effects (errors, failures)
  - fault tolerant design techniques
  - evaluation of fault-tolerant systems
- balance
  - concepts, underlying principles
  - applications

Focus-Overview

- Introduction
  - definition of fault tolerance, applications
- Fundamentals of dependability
  - dependability attributes : reliability, availability, safety
  - dependability impairments : faults, errors, failures
  - dependability means
- Dependability evaluation techniques
  - common measures: failure rate, MTTF, MTTR
  - reliability block diagrams
  - Markov processes

Focus-Overview

- Fault Tolerant design techniques
  - space redundancy
    - hardware redundancy
    - information redundancy
    - software redundancy
  - time redundancy

Textbook

- E. Dubrova, **Fault-Tolerant Design**, Springer, 2013.